Moving sentinel world: communication contrast depends on field of view

The hidden-state/action relation for this extension is summarized in the categorical loop schematic fig. 8; the results below remain the executed moving-world comparisons,

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Moving sentinel world (part 2)

not a claim that the schematic's full loop is present in every flat belief-sharing study.

Moving sentinel world (part 3)

The static sentinel world lets every agent observe the same shared latent, so belief sharing is a refinement rather than a requirement. To stress the *necessity* of communication we add movement and **disjoint** fields of view.

Moving sentinel world (part 4)

The world is a linear grid of 4 cells holding a single binary threat — left half (state 0) or right half (state 1). The 2 sentinels start at evenly tiled positions and each observe a half-open window of cells, so in the default setup agent 0 watches the left half and agent 1 the right half: their views do not overlap.

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Moving sentinel world (part 5)

Each agent's likelihood is a confident, signed presence reading for the half it can see, and three control paths (stay / left / right) let it reposition. The expected-free-energy policy scores each candidate move by the expected posterior entropy after one observation and takes the most information-seeking step.

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Moving sentinel world (part 6)

We run 960 trials of 6 steps each under three conditions: *isolated* (random moves, no sharing), *communicating* (random moves plus a log-linear-pool consensus each step), and *EFE-guided* (information-seeking moves plus the same sharing).

Moving sentinel world (part 7)

The measured consensus accuracies are 0.999 (isolated), 0.977 (communicating), and 0.978 (EFE-guided), with a communicating free-energy gap of -0.528 nats relative to the isolated baseline (negative:

Moving sentinel world (part 8)

no free-energy advantage over isolated in this binary-complement regime of logically complete half-views) (fig. 29).

Moving sentinel world (part 9)

Across 128 independent seeds the EFE-guided accuracy is 0.983 (95 % CI 0.982–0.984), the communicating (random-moves + sharing) accuracy is 0.982 (95 % CI 0.982–0.983), and the isolated accuracy is 0.999 (95 % CI 0.999–0.999).

Moving sentinel world (part 10)

In this binary-complement regime the isolated condition is in fact significantly *higher* on accuracy than the EFE-guided sharing condition — their 95 % intervals do not overlap — and the EFE-vs-isolated accuracy contrast yields Wilcoxon signed-rank $p = 9.27 \times 10^{-23}$ (significant; isolated higher),

Moving sentinel world (part 11)

effect size $r = 1.000$ (large). Sharing is therefore not merely unnecessary in this regime; it costs a small but reliable amount of accuracy.

Moving sentinel world (part 12)

Nor does sharing lower free energy here: the EFE free-energy gap (isolated surprise minus the EFE-guided condition's surprise) is -0.368 nats (95 % CI -0.384—0.353) — negative, so the pooled consensus is slightly *more* surprised by the true state than the isolated baseline, because a single agent's view already suffices.

Moving sentinel world (part 13)

The accuracy case for *necessity* is therefore made only in the larger-state-space disjoint-FOV extension below, not by these binary-complement numbers.

Moving sentinel world (part 14)

We report these numbers as measured, not assumed. The binary world carries a logical complement: ruling out one's own half implies the other, so a single agent's "not detected" still carries information about the global state, and an isolated agent is not strictly blind.

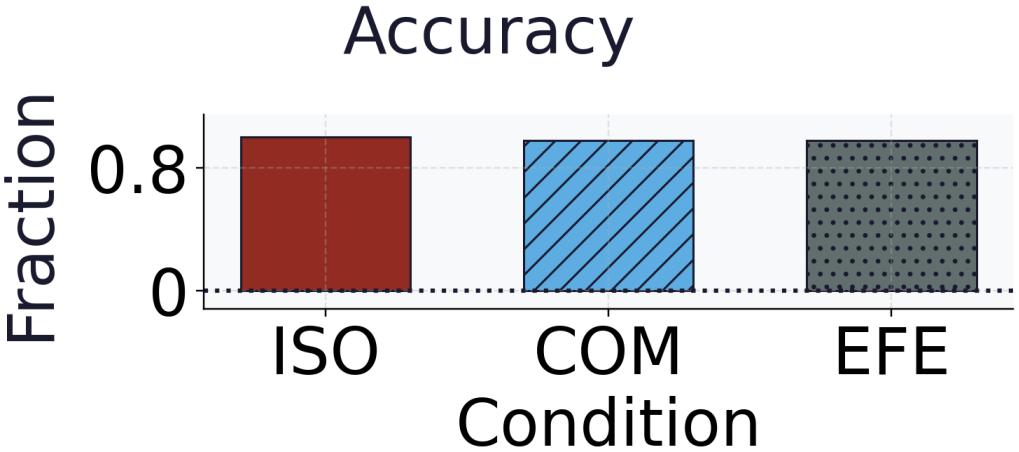
Moving sentinel world (part 15)

By design, the intended *cannot-decide-alone* regime is the high-noise, few-step corner where one sensor's evidence cannot overcome the flat prior; there belief sharing is meant to fuse the two complementary views into a decisive consensus. That regime is not separately measured here — the construction,

Moving sentinel world (part 16)

the three actions, the EFE rule, and the exact condition protocol are detailed in the supplement (sec. 17.8).

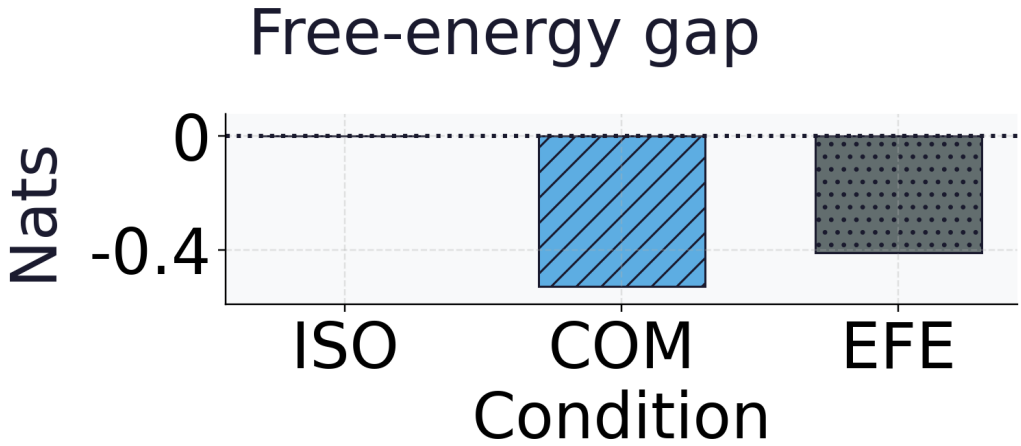
Moving sentinel world (part 17)



Moving sentinel world (part 18)

Accuracy (fraction): ISO +0.999; COM +0.9771; EFE +0.9781.

Moving sentinel world (part 19)

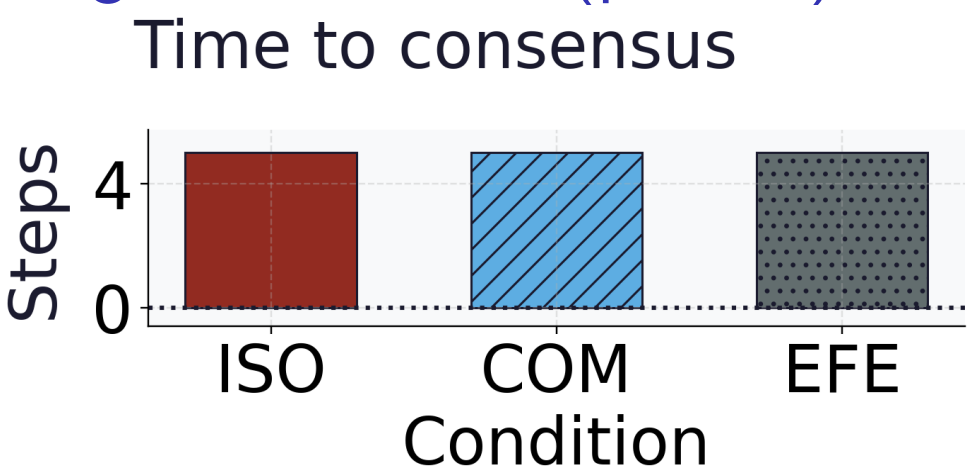


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Moving sentinel world (part 20)

Free-energy gap (nats): ISO +0; COM -0.5285; EFE -0.4099.

Moving sentinel world (part 21)



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Moving sentinel world (part 22)

Time to consensus (steps): ISO +5.002; COM +5; EFE +5.001.

Moving sentinel world (part 23)

Accuracy gains over isolated: COM -0.022; EFE -0.021.

Moving sentinel world (part 24)

ISO / plain: isolated; COM / forward hatch: communicating; EFE / dots: EFE-guided. Different metric units have separate axes.

Moving sentinel world (part 25)

Declared moving-world summary. No interval is added to these scalar bars. Ordered steps and within-run observations do not become independent replications.

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Disjoint field-of-view extension

To test whether communication is necessary (not merely beneficial) when observations are non-overlapping, we extend Study 5 to 3 agents each observing a 2-position disjoint window of a 6-position state space (chance-level accuracy 0.167).

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Disjoint field-of-view... (part 2)

Isolated agents achieve mean accuracy 0.35 — above chance but far from decisive, since no single agent can infer the global state from a partial window alone. Communicating agents pool complementary beliefs to reach 0.55 (gap 0.19).

Disjoint field-of-view… (part 3)

This is now a powered result, not an illustrative point estimate. Across 128 independent seeds the isolated accuracy is 0.326 (95% CI 0.320–0.332) and the communicating accuracy is 0.493 (95% CI 0.487–0.499), both clearing the 0.167 chance baseline — isolated agents are *not* at chance,

Disjoint field-of-view... (part 4)

since a partial FOV plus majority voting still carries some signal.

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Disjoint field-of-view… (part 5)

The paired Wilcoxon signed-rank test (communicating vs. isolated, matched by seed) gives $p = 9.35 \times 10^{-23}$, effect size $r = 1.000$ (large): communicating beats isolated on every one of the 128 seeds. This agreement saturates the rank-biserial effect at its upper bound;

Disjoint field-of-view... (part 6)

it does not turn the reported Wilcoxon signed-rank p-value into an exact paired sign-test probability. The paired mean gap and its interval quantify the accuracy contrast in its native units.

Disjoint field-of-view... (part 7)

Given isolated performance is above chance, the precise claim is not that communication is *logically* necessary for any signal at all, but that it is necessary to approach the communicating-level accuracy under fully disjoint observations: the gap between the two conditions is significant, large,

Disjoint field-of-view… (part 8)

and reproducible, unlike the binary-complement contrast above.

Disjoint field-of-view... (part 9)

We separately quantify EFE-guided navigation rather than asserting an unquantified "widens the gap" effect. In a matched but smaller-scale disjoint-FOV movement-policy comparison (2 agents, 4-position binary-state grid, belief sharing active in both arms),

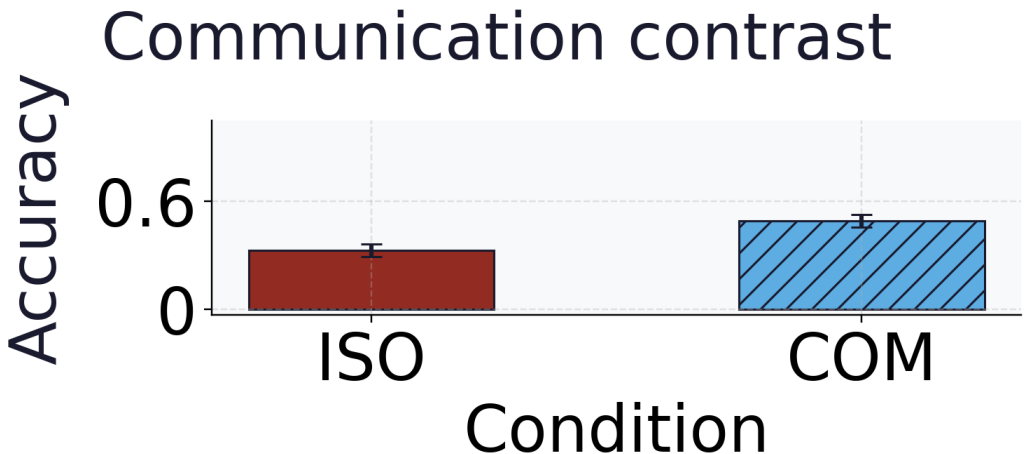
Disjoint field-of-view... (part 10)

EFE-guided accuracy is 0.975 versus 0.977 for random movement ($p = 0.1046$, negligible effect). This comparison did not resolve an accuracy difference in the configured near-ceiling regime with belief sharing active; it does not establish equivalence of the movement policies.

Disjoint field-of-view… (part 11)

We report this as the null result it is rather than claiming an unmeasured EFE benefit. fig. 30 summarizes the necessity result.

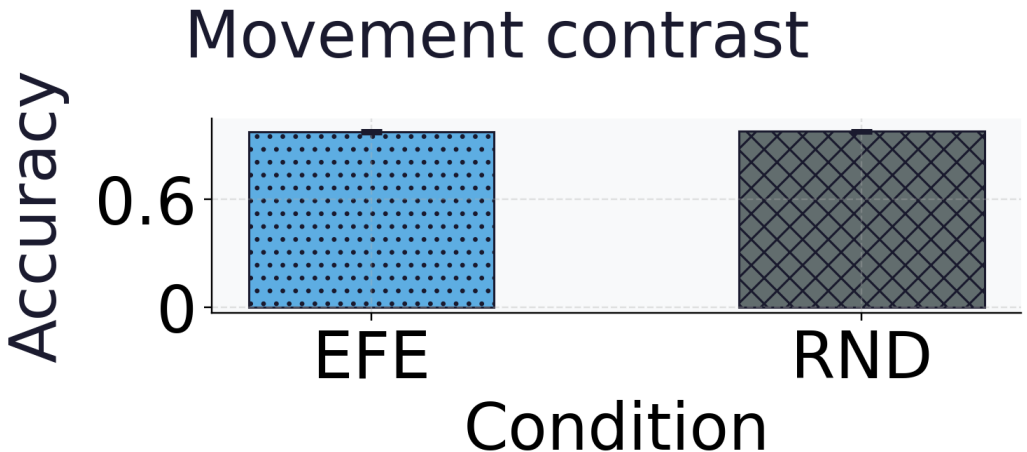
Disjoint field-of-view... (part 12)



Disjoint field-of-view… (part 13)

ISO mean 0.3262, SD 0.03472; COM mean 0.4928, SD 0.0349. COM−ISO = +0.1666.

Disjoint field-of-view... (part 14)



Disjoint field-of-view... (part 15)

EFE mean 0.9749, SD 0.01134; RND mean 0.9768, SD 0.01019. EFE−RND = -0.001953.

Disjoint field-of-view… (part 16)

ISO: isolated; COM: communicating; EFE: EFE-guided; RND: random movement. Hatches preserve condition identity.

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Disjoint field-of-view… (part 17)

Whiskers show standard deviations across configured seeds, not confidence intervals. Accuracy is a fraction correct.

Disjoint field-of-view… (part 18)

Separate communication and movement-policy contrasts in the declared disjoint-view worlds. A near-ceiling movement contrast does not imply a general navigation advantage.

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